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問題1

(1) $\tilde{a}_{11} = (-1)^{1+1}\begin{vmatrix} 2 & 2 \\ 0 & 1 \end{vmatrix} = 2$

$\tilde{a}_{31} = (-1)^{3+1}\begin{vmatrix} 1 & -1 \\ 2 & 2 \end{vmatrix} = 4$

(2) $|A| = a_{11}\cdot\tilde{a}_{11} + a_{21}\tilde{a}_{21} + a_{31}\cdot\tilde{a}_{31}$　　$\tilde{a}_{21} = (-1)^{2+1}\begin{vmatrix} 1 & -1 \\ 0 & 1 \end{vmatrix} = -1$

$= 2\times 2 + 0\times(-1) + 1\times 4$

$= 8$

(3) $A^2 = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 2 & 2 \\ 1 & 0 & 1 \end{bmatrix}\begin{bmatrix} 2 & 1 & -1 \\ 0 & 2 & 2 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 4 & -1 \\ 2 & 4 & 6 \\ 3 & 1 & 0 \end{bmatrix}$

$A^3 = \begin{bmatrix} 3 & 4 & -1 \\ 2 & 4 & 6 \\ 3 & 1 & 0 \end{bmatrix}\begin{bmatrix} 2 & 1 & -1 \\ 0 & 2 & 2 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 11 & 4 \\ 10 & 10 & 12 \\ 6 & 5 & -1 \end{bmatrix}$

$A^5 = A^2\cdot A^3 = \begin{bmatrix} 3 & 4 & -1 \\ 2 & 4 & 6 \\ 3 & 1 & 6 \end{bmatrix}\begin{bmatrix} 5 & 11 & 4 \\ 10 & 10 & 12 \\ 6 & 5 & -1 \end{bmatrix} = \begin{bmatrix} 49 & 68 & 61 \\ 86 & 92 & 50 \\ 61 & 73 & 18 \end{bmatrix}$

$|A^5| = \begin{vmatrix} 49 & 68 & 61 \\ 86 & 92 & 50 \\ 61 & 73 & 18 \end{vmatrix} = 45056$

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問題2:

(1) $\tilde{a}_{11} = (-1)^{1+1}\begin{vmatrix} 4 & 1 \\ 5 & 2 \end{vmatrix} = 3$ $\tilde{a}_{21} = (-1)^{2+1}\begin{vmatrix} 2 & -1 \\ 5 & 2 \end{vmatrix} = -9$

$\tilde{a}_{31} = (-1)^{3+1}\begin{vmatrix} 2 & -1 \\ 4 & 1 \end{vmatrix} = 6$ $\tilde{a}_{12} = (-1)^{1+2}\begin{vmatrix} 3 & 1 \\ -2 & 2 \end{vmatrix} = -8$

$\tilde{a}_{13} = (-1)^{1+3}\begin{vmatrix} 3 & 4 \\ -2 & 5 \end{vmatrix} = 23$ $\tilde{a}_{22} = (-1)^{2+2}\begin{vmatrix} 1 & -1 \\ -2 & 2 \end{vmatrix} = 0$

$\tilde{a}_{23} = (-1)^{2+3}\begin{vmatrix} 1 & 2 \\ -2 & 5 \end{vmatrix} = -9$ $\tilde{a}_{32} = (-1)^{3+2}\begin{vmatrix} 1 & -1 \\ 3 & 1 \end{vmatrix} = -4$

$\tilde{a}_{33} = (-1)^{3+3}\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = -2$

$$\tilde{A} = \begin{bmatrix} 3 & -8 & 23 \\ -9 & 0 & -9 \\ 6 & -4 & -2 \end{bmatrix}$$

(2) $|A| = \begin{vmatrix} 1 & 2 & -1 \\ 3 & 4 & 1 \\ -2 & 5 & 2 \end{vmatrix} = 1\times4\times2 + (-2)\times2\times1 + 3\times5\times(-1) - (-2)\times4\times(-1)$
$- 2\times3\times2 - 1\times1\times5 = -36$

(3) $$A^{-1} = \frac{1}{|A|}\tilde{A} = \begin{bmatrix} -\frac{1}{12} & \frac{2}{9} & -\frac{23}{36} \\ \frac{1}{4} & 0 & \frac{1}{4} \\ -\frac{1}{6} & \frac{1}{9} & \frac{1}{18} \end{bmatrix}$$

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問題3

$$\begin{bmatrix} 2 & 3 & -5 \\ 1 & -1 & 1 \\ 3 & -6 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ -7 \end{bmatrix}$$

$$x = \frac{\begin{vmatrix} 3 & 3 & -5 \\ 0 & -1 & 1 \\ -7 & -6 & 2 \end{vmatrix}}{\begin{vmatrix} 2 & 3 & -5 \\ 1 & -1 & 1 \\ 3 & -6 & 2 \end{vmatrix}} = \frac{26}{26} = 1$$

$$y = \frac{\begin{vmatrix} 2 & 3 & -5 \\ 1 & 0 & 1 \\ 3 & -7 & 2 \end{vmatrix}}{\begin{vmatrix} 2 & 3 & -5 \\ 1 & -1 & 1 \\ 3 & -6 & 2 \end{vmatrix}} = \frac{52}{26} = 2$$

$$z = \frac{\begin{vmatrix} 2 & 3 & 3 \\ 1 & -1 & 0 \\ 3 & -6 & -7 \end{vmatrix}}{\begin{vmatrix} 2 & 3 & -5 \\ 1 & -1 & 1 \\ 3 & -6 & 2 \end{vmatrix}} = \frac{26}{26} = 1$$

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問題4:

$\{a, b, c, d\}$ $ma+nb+pc+qd=0$ とする.

$$\begin{cases} m+3q=0 \\ 2m-2n+2p=0 \\ m-2n+2p-3q=0 \\ -2m+p+6q=0 \end{cases} \Rightarrow \begin{bmatrix} 1 & 0 & 0 & 3 \\ 2 & -2 & 2 & 0 \\ 1 & -2 & 2 & -3 \\ -2 & 0 & 1 & 6 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 15 \\ 0 & 0 & 1 & 12 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\downarrow$$

$$\begin{cases} m=-3q \\ n=-15q \\ p=-12q \end{cases}$$

$\{a, b, c\}$は1次独立なベクトルを得る.

残りのベクトル$\{d\}$

$$d = -3a+15b+12c$$

問題5

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$\{a, b\}$ は1次独立であるので.

$\{a, b\}$: $x_1 a + x_2 b = 0$ $(x_1 = x_2 = 0)$ である.

$c = 2a + b$, $d = a + 2b$. とすると.

$$y_1 c + y_2 d = 0 \Rightarrow (2y_1 + y_2) a + (y_1 + 2y_2) b = 0$$

すなわち, $\begin{cases} 2y_1 + y_2 = x_1 = 0 \\ y_1 + 2y_2 = x_2 = 0 \end{cases} \Rightarrow \begin{cases} y_1 = 0 \\ y_2 = 0 \end{cases}$

つまり. $\{c, d\}$ は1次独立である.

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問題6.

$a = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \in W$ とおくと

$3a = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$ をすると $3+3 \neq 3 \times 3$ ので

$3a \notin W$

Wが部分空間でない.